Palak Deb Patra

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EDUCATION

National Institute of Technology, Rourkela — **Dec 2021 – Jun 2025**
Bachelor of Technology in **Computer Science & Engineering** | **CGPA: 8.52/10** — **Rourkela, India**

PROFESSIONAL EXPERIENCE

NatWest Group | **Software Engineer** — **Jul 2025 – Present**
Python, FastMCP, Azure, Microsoft Copilot Studio, Power Platform — **Bangalore, India**

Project: Colleague Learning Assistant (CLA)

- Built **end-to-end AI observability** with **OpenTelemetry** and **Azure Application Insights**, enabling request, tool-call and conversation tracing through a unified ID, with **8 failure categories** and **11 environment-specific settings**.
- Developed **131 automated tests**, **11 Kusto queries** and a **5-tab Application Insights dashboard** to validate telemetry, investigate failures and monitor agent activity.
- Configured **SSO** with **PingFederate** and **Microsoft Entra ID**, using a **three-step fallback** to identify employees and prevent incorrect user identification.
- Integrated **12 months of Workday learning records** to personalise course recommendations and built an **Azure Blob Storage pipeline** for GitLab metrics with change detection and fallback to the last valid copy.
- Redesigned **3 Microsoft Teams widgets** for Learning Courses, ECDF progress and GitLab metrics to follow NatWest branding and resolve rendering issues.

Project: ChLoE

- Worked on migrating deployments from **Power Platform pipelines** to **GitLab CI/CD** and **Ansible Automation Platform** in line with bank security and release requirements.
- Reduced **agent hallucinations** by improving Power Platform FAQ knowledge sources and validating revised responses in SIT and UAT, with **23 of 26 test cases passing**.
- Fixed **duplicate agent responses** after deployments and corrected the Microsoft Teams **"Start over" trigger** to clear chat history and restart conversations.

NatWest Group | **Software Engineering Intern** — **May 2024 – Jul 2024**
Python, GraphQL, REST APIs, Snowflake — **Gurgaon, India**

Project: OSPO Metrics

- Collected organisation-wide **DevOps metrics** through **GitLab GraphQL and REST APIs** and automated data extraction and CSV exports **in Python**, reducing manual processing time by **28%**.
- Loaded **16 datasets** into **Snowflake** and developed a unified data model for interactive **ThoughtSpot dashboards**.

PERSONAL PROJECTS

CodeShift | Python, LangGraph, Ollama, TypeScript | GitHub — **Aug 2026**

- Built a **six-agent system** using LangGraph and Ollama to migrate Python projects to TypeScript in dependency order, with **mypy**, **tsc** and differential tests for verification.
- Evaluated a **31-module accounting application** across **197 functions and methods**, confirming matching behaviour for **15 modules** under the generated tests.

Memex | Python, LangGraph, ChromaDB, Ollama, Streamlit | GitHub — **Aug 2026**

- Built a **local RAG system** with cited answers, OCR and automatic index updates, combining **dense retrieval, BM25 and cross-encoder reranking**; indexed **3,050 files** into **42,743 chunks** in about **15 minutes** on my local GPU.
- Evaluated retrieval on **42 questions** from a separate **23-document corpus**, achieving **97.6% Hit@1**, **0.988 MRR** and **100% Recall@5**.

TECHNICAL SKILLS

Languages	Python, Java, C++, JavaScript, TypeScript, SQL
AI & ML	PyTorch, TensorFlow, scikit-learn, OpenCV, spaCy, NLTK,
LLMs & Agents	LangGraph, LangChain, Hugging Face, Ollama, FastMCP, Claude, Codex
Web & Data	React, Angular, Streamlit, FastAPI, REST APIs, GraphQL, Snowflake, Oracle, MySQL, ChromaDB
Cloud & Tools	AWS, Azure, NVIDIA CUDA, Git, GitLab CI/CD, Ansible, Jupyter

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2023**.
- Solved **800+** LeetCode problems with a peak rating of **1795**, ranking in the **top 8%** globally.